A common-affine residual class and a smooth obstruction for conditional entropy-power concavity

Candidate partial result for arXiv:2501.10309

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Status. Candidate partial result, judged likely valid. The packet gives a non-Gaussian positive class and disproves unrestricted concavity, but it does not characterize all admissible classes and does not address the Ball–Nayar–Tkocz log-concave conjecture.

1 Source question

Fradelizi, Gavalakis, and Rapaport define, for independent random vectors $X, Y \in \mathbb{R}^n$, the map

$$F(\lambda) = \frac{\mathsf{N}_n(\sqrt{\lambda}X + \sqrt{1-\lambda}Y)^n}{\mathsf{N}_{n-1}(\sqrt{\lambda}X' + \sqrt{1-\lambda}Y')^{n-1}}, \qquad 0 \le \lambda \le 1, \tag{1}$$

where X', Y' denote the first $n-1$ coordinates and $\mathsf{N}_d(W) = \exp(2h(W)/d)$ [1]. In Remark 9, equation (18), page 7, they ask for which classes of random variables this map is concave.

$\square$

Remark 9. *Notice that by defining the map:*

$$f : \lambda \mapsto \frac{N\left(\sqrt{\lambda}X + \sqrt{1-\lambda}Y\right)^n}{\left(N_{n-1}(\sqrt{\lambda}X^{n-1} + \sqrt{(1-\lambda)}Y^{n-1})\right)^{n-1}}, \tag{18}$$

Theorem 6 states that

$$f(\lambda) = f(\lambda \cdot 1 + (1-\lambda) \cdot 0) \ge \lambda f(1) + (1-\lambda) f(0).$$

Ball, Nayar and Tkocz [15] conjectured that the map $\lambda \mapsto h(\sqrt{1-\lambda}X + \sqrt{\lambda}Y)$ is concave for i.i.d. log-concave X, Y in dimension 1. A partial result in a different direction was given in [16], where it shown that this entropy-map is concave for Gaussian (scale) mixtures.

In view of the above conjecture and the partial answer for Gaussian mixtures it is natural to ask for which class of random variables (if any) is the map defined in (18) concave.

Note that, in contrast to these works, when restricting to $d = 1$, we are asking for concavity of the entropy power, rather than the entropy itself, which is stronger, since concavity implies log-concavity.

When one of the two random vectors is Gaussian, by homogeneity, concavity of the map defined above is equivalent to concavity of

$$f : \lambda \mapsto \frac{N\left(X + \sqrt{\lambda}Z\right)^n}{\left(N_{n-1}(X^{n-1} + \sqrt{\lambda}Z^{n-1})\right)^{n-1}},$$

Figure 1: The source question in Remark 9 on page 7 of arXiv:2501.10309v1.

The source proves the endpoint-chord inequality $F(\lambda) \ge \lambda F(1) + (1-\lambda)F(0)$, which is weaker than concavity at arbitrary pairs of parameters.

2 Idea of the proof

The quotient in (1) is not an opaque ratio: it is exactly the exponential of twice a conditional entropy. If the last coordinates of X and Y have the same affine regression on their first blocks and the residuals are independent, conditioning removes the common affine term and the problem becomes the ordinary one-dimensional entropy-power problem for the two residuals. Costa's theorem then gives a positive result whenever one residual is Gaussian, even though the first $n-1$ coordinates can be wholly non-Gaussian.

The same reduction exposes an obstruction. Smooth a Rademacher variable by a very narrow Gaussian. At unequal coefficients, four discrete modes remain separated; at equal coefficients, the two middle modes collide. In the small-noise limit this lowers discrete entropy by $\frac{1}{2}\log 2$, and hence lowers entropy power by a factor tending to $1/2$. Midpoint concavity therefore fails. The resulting random vectors are iid and have everywhere positive smooth densities. What remains open is a broader structural classification, especially under log-concavity.

3 Conditional entropy reformulation

Write $W = (W', W_n) \in \mathbb{R}^{n-1} \times \mathbb{R}$. Whenever the displayed entropies are finite,

$$\frac{\mathsf{N}_n(W)^n}{\mathsf{N}_{n-1}(W')^{n-1}} = \exp\big(2h(W) - 2h(W')\big) = \exp\big(2h(W_n \mid W')\big). \tag{2}$$

Thus F is a conditional entropy power.

4 A non-Gaussian positive class

Theorem 1 (Common-affine independent residuals)**.** *Let $X = (X', X_n)$ and $Y = (Y', Y_n)$ be independent random vectors in $\mathbb{R}^n$. Assume that, for some $a \in \mathbb{R}^{n-1}$,*

$$X_n = a^T X' + U, \qquad Y_n = a^T Y' + V,$$

where $U \perp\!\!\!\perp X'$ and $V \perp\!\!\!\perp Y'$. Assume the finite-entropy and finite-second- moment hypotheses needed below. Then

$$F(\lambda) = \mathsf{N}_1(\sqrt{\lambda}U + \sqrt{1-\lambda}V). \tag{3}$$

In particular, if one of U, V is a nondegenerate Gaussian, then F is concave on $[0, 1]$.

Proof. Put $W_\lambda = \sqrt{\lambda}X + \sqrt{1-\lambda}Y$ and $R_\lambda = \sqrt{\lambda}U + \sqrt{1-\lambda}V$. The hypotheses and the independence of X and Y imply

$$(W_\lambda)_n = a^T (W_\lambda)' + R_\lambda, \qquad R_\lambda \perp\!\!\!\perp (W_\lambda)'.$$

Translation by the conditioned value and independence give $h((W_\lambda)_n \mid (W_\lambda)') = h(R_\lambda)$. Equation (2) proves (3).

Suppose, for definiteness, that $V = \tau Z$ with Z standard Gaussian. Costa's entropy-power theorem states that $G(t) = \mathsf{N}_1(U + \sqrt{t}Z)$ is concave for $t \geq 0$ [2]. For $0 < \lambda \leq 1$, homogeneity of entropy power gives

$$F(\lambda) = \lambda G\left(\frac{\tau^2(1-\lambda)}{\lambda}\right).$$

The perspective $(x, y) \mapsto xG(y/x)$ of a concave function is concave on $x > 0$, $y \geq 0$. Restrict it to the affine segment $(x, y) = (\lambda, \tau^2(1-\lambda))$ and pass to the endpoints by continuity. The case in which U is Gaussian is symmetric. $\square$

The first blocks X' and Y' in Theorem 1 are arbitrary; in particular, the full vectors need not be Gaussian or Gaussian mixtures.

5 Smooth iid failure of unrestricted concavity

We use the following elementary small-noise fact.

Lemma 1 (Finite mixtures at small noise)**.** *Let A be a finite-valued real random variable with distinct support points, and let Z be an independent standard Gaussian. Then*

$$h(A + \sigma Z) = h(\sigma Z) + H(A) + o(1) \qquad (\sigma \downarrow 0),$$

where H is discrete Shannon entropy.

Proof. The mutual-information identities give

$$h(A + \sigma Z) - h(\sigma Z) = I(A; A + \sigma Z) = H(A) - H(A \mid A + \sigma Z).$$

Let $\delta > 0$ be the minimum distance between distinct support points of A. Nearest-neighbor decoding has error probability at most $\Pr\{|\sigma Z| \geq \delta/2\}$, which tends to zero. Fano's inequality therefore gives $H(A \mid A + \sigma Z) \to 0$. $\square$

Theorem 2 (Smooth iid mode-collision obstruction)**.** *For every $n \geq 1$ there are iid random vectors $X, Y \in \mathbb{R}^n$ with everywhere positive C^∞ densities and finite moments of all orders for which the map F in* (1) *is not concave. More explicitly, let S, T be independent Rademacher variables, let Z_1, Z_2 be independent standard Gaussians, and set*

$$U_\sigma = 4S + \sigma Z_1, \qquad V_\sigma = 4T + \sigma Z_2.$$

For all sufficiently small $\sigma > 0$, these residuals, together with arbitrary iid independent Gaussian first blocks, give

$$F_\sigma(1/2) < \frac{F_\sigma(1/4) + F_\sigma(3/4)}{2}.$$

Proof. Take $X = (G, U_\sigma)$ and $Y = (H, V_\sigma)$, where G, H are iid standard Gaussian vectors in $\mathbb{R}^{n-1}$, independent of all scalar variables; for $n = 1$ omit the first blocks. The vectors X, Y are iid with positive smooth densities. Theorem 1's reduction (which does not require a Gaussian residual for the identity itself) yields

$$F_\sigma(\lambda) = \mathsf{N}_1(\sqrt{\lambda} U_\sigma + \sqrt{1-\lambda} V_\sigma).$$

Because U_σ, V_σ are iid, $F_\sigma(1/4) = F_\sigma(3/4)$.

At $\lambda = 1/4$, Gaussian stability gives

$$\sqrt{\lambda} U_\sigma + \sqrt{1-\lambda} V_\sigma \stackrel{d}{=} A + \sigma Z, \qquad A = 2S + 2\sqrt{3}\, T,$$

where Z is standard Gaussian. The four values of A are distinct and equiprobable, so $H(A) = 2 \log 2$. At $\lambda = 1/2$ the corresponding discrete part is

$$B = 2\sqrt{2}(S + T),$$

whose three values have probabilities $1/4, 1/2, 1/4$; hence $H(B) = \frac{3}{2} \log 2$. Lemma 1 now gives

$$\frac{F_\sigma(1/2)}{F_\sigma(1/4)} = \exp(2[h(B + \sigma Z) - h(A + \sigma Z)]) \longrightarrow \exp(-\log 2) = \frac{1}{2}.$$

Thus $F_\sigma(1/2) < F_\sigma(1/4) = F_\sigma(3/4)$ for all sufficiently small σ, contradicting midpoint concavity. $\square$

6 Computational sanity check

The script `code/explore_gaussian_mixtures.py` numerically integrates the exact four-component Gaussian-mixture density. With component means ± 4 and component standard deviation $\sigma = 0.1$, an 80,001-point grid gave

$$(F(1/4), F(1/2), F(3/4)) \approx (2.7327149513, 1.3663574756, 2.7327149513).$$

This is only a check. The proof of Theorem 2 is the small-noise argument above and does not rely on numerical quadrature or on proving that 0.1 lies below an explicit threshold.

7 Verification and limitations

An adversarial step check is supplied in `verification.md`. The core proof is self-contained apart from Costa's theorem in the positive direction. The negative theorem uses only the entropy chain rule, elementary decoding, and Fano's inequality.

The result is partial. It neither classifies all pairs (X, Y) nor resolves the log-concave two-weight entropy conjecture cited by the source. The negative examples are smooth and symmetric but non-log-concave for small σ.

Bounded novelty check. Searches made on 19 July 2026 used the exact source phrase and combinations of entropy power concavity, Rademacher Gaussian mixture, iid, and Costa. The search also checked the generalized-EPI counterexample paper of Madiman–Nayar–Tkocz and the centered Gaussian scale-mixture paper of Eskenazis–Gavalakis [3, 4]. Those works concern related entropy or multiweight phenomena. No exact match was found for the conditional quotient reduction or the two-weight smooth mode-collision proof above. This was a bounded search, not an exhaustive novelty certification.

Human-review recommendation. Send to a specialist reviewer. The reviewer should check the preferred regularity formulation of Costa's theorem, the independence bookkeeping in Theorem 1, and whether a later version of the source already records the same common-affine residual corollary.

References

[1] M. Fradelizi, L. Gavalakis, and M. Rapaport, *Entropic versions of Bergström's and Bonnesen's inequalities*, arXiv:2501.10309v1, 2025.

[2] M. H. M. Costa, *A new entropy power inequality*, IEEE Trans. Inform. Theory 31 (1985), no. 6, 751–760.

[3] M. Madiman, P. Nayar, and T. Tkocz, *Two remarks on generalized entropy power inequalities*, arXiv:1904.02314, 2019.

[4] A. Eskenazis and L. Gavalakis, *On the entropy and information of Gaussian mixtures*, Mathematika 70 (2024), e12246; arXiv:2308.15997.